



KTU NOTES

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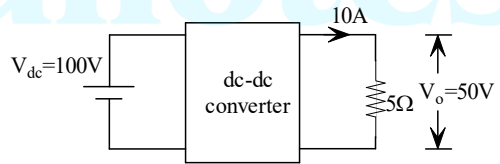
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used for amplification and it is operated in active region. But, power BJT is used in OFF) hence it is operated in saturation region or cut-off region. When BJT is in saturation region, power dissipation is more ($=V_{CE} \times I_C$). When a BJT is operated in saturation region, power dissipation is less. ON state (during saturation, V_{CEsat} is very low) and OFF state (I_C is very low) losses are minimum.

In linear voltage regulator, line frequency transformer which is heavy & inefficient. In switched mode regulator, high frequency ferrite core transformer which is light & efficient.

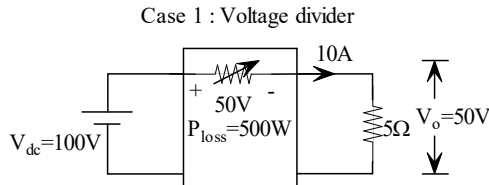
For example,

Consider a simple dc-dc converter. A 50V dc is required across a 5Ω resistor. Load current is 10A (dc load current is 10A).

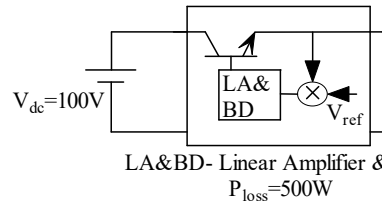


The different ways of getting a reduced dc voltage from a fixed dc voltage are:

Case 1 : A voltage divider is used. The voltage drop across the variable resistor is 50V. The power dissipated is 500W.

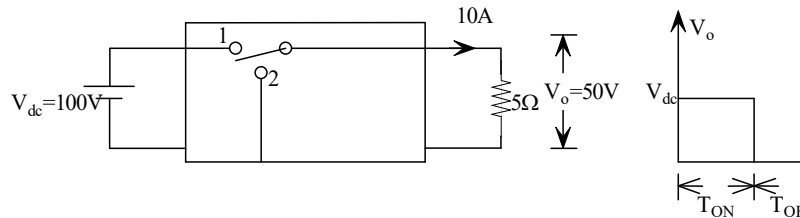


Case 2 : Linear series regulator



Case 2 : A linear voltage regulator is used. The transistor is operating in active region. The voltage drop across the transistor is 50V and the power dissipated in the transistor is 500W.

Case 3



- constant)
- ii) Inverters – fixed DC to variable AC (variable V or f) (Average value i sine wave)
- iii) Choppers – Fixed DC to variable DC
- iv) Voltage controllers – Fixed ac to variable ac (Variable V; fixed f)
- v) Cycloconverters & Matrix converters – Fixed ac to variable ac (variable V; fixed f)

Power Semiconductor devices :

- i) Uncontrolled devices – Power diode
- ii) Semiconrolled devices (controlled turn-on and uncontrolled turn-off)
Rectifier (SCR)
- iii) Fullycontrolled devices (controlled turn-on and controlled turn-off) – P MOSFET, GTO (Gate Turn Off thyristor), IGBT (Insulated Gate B IGBT, IGCT (Integrated Gate Commutated Thyristor)

COMPARISON OF POWER SEMICONDUCTOR DEVICES

Device	Rated voltage	Rated current	Rated frequency	Rated power	Power
Thyristor (SCR)	6kV	3.5kA	500Hz	100's MW	1
BJT	1.2kV	400A	10kHz	1MW	1
MOSFET	1kV	50A	1MHz	100kW	3
IGBT	1.2kV	400A	20kHz	100's kW	3
GTO	4.5kV	3kA	2kHz	10's MW	3

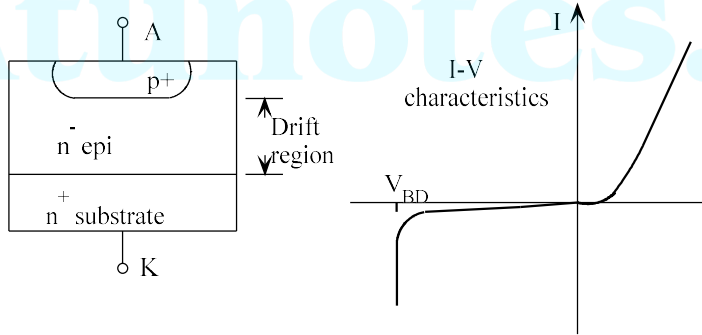
IDEAL SWITCHES

Properties of ideal switches:

- i) When switch is OFF (open); $I_s = 0$ and should be able to withstand any voltage $(-\infty \leq V_{sw(OFF)} \leq +\infty)$
- ii) When switch is ON, $V_{sw(ON)}=0$ and it is capable of passing any current through it
- iii) Power dissipated in the switch when ON or OFF is zero (Conduction loss = 0)
- iv) Switch is turned ON and OFF instantaneously i.e. $t_{ON} = t_{OFF} = 0$.
- v) Switching losses during turn ON and turn OFF are zero.

POWER DIODE

Basic structure of a power diode is shown below. It consists of a heavily doped n^- epitaxial layer of specified thickness. Formed by diffusing in a heavily doped p-type region that forms the anode of the diode. The sectional area of the diode will vary according to the amount of total current it has to carry.



The n^- epitaxial layer (lightly doped) (drift region) is not found in ordinary low power diodes. The depletion layer of reverse biased p^+n^- junction. Width of the drift region decides the reverse voltage capability. This may be quite wide at large reverse voltages. Relatively long drift region increases the ON state drop which increases the power loss during conduction.

In the forward direction, the characteristics of power diode and ordinary diode are different. In a power diode the current grows linearly with the forward-bias voltage rather than exponentially. The ohmic drops in a power diode create ohmic drops that mask the exponential i-v characteristics. The lightly doped drift region accounts for part of this ohmic resistance.

During reverse bias, only a leakage current flows upto V_{BD} where avalanche breakdown occurs; large current flows; large current at large voltage leads to high power dissipation; the device is destroyed.

Diodes may be punch through diode or non-punch through diode.

In non-punch through diode, width of depletion layer at breakdown is less than the width of the drift region. Depletion layer does not reach n^+ substrate.

Power diode requires finite time to switch from the blocking state to ON state and vice versa.

Reverse recovery charge, $q_{rr} = I_{rr} \times \frac{t_{rr}}{2}$

HEAT SINK :

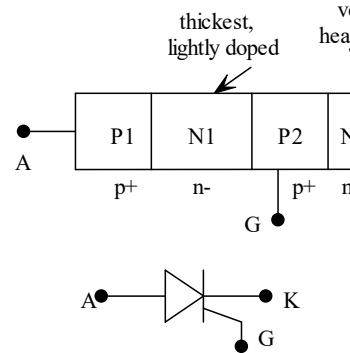
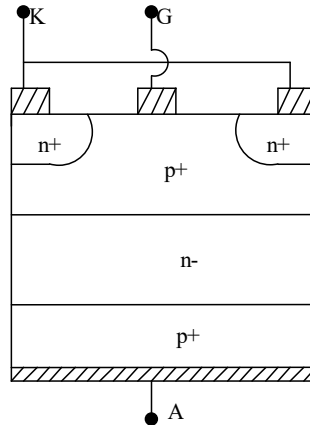
Due to ON state and switching losses, heat is generated within the device transferred from the device to a cooling medium to maintain the operating junction temperature within the specified range. The device is usually mounted on a heat sink (usually aluminum).

SPECIFICATIONS OF POWER DIODE: (R – reverse; R – Repetitive; M – Peak)

Surge)

- i) Average forward current $I_{F(AV)}$
- ii) Reverse blocking voltage V_{RRM} (Reverse Repetitive Peak Voltage)
- iii) ON state voltage V_F
- iv) Reverse recovery time t_{rr}
- v) Surge current rating I_{FSM}
- vi) i^2t rating

SILICON CONTROLLED RECTIFIER (SCR)



SCR has 3 modes of operation i) forward blocking mode; 2) forward conduction mode; 3) reverse blocking mode

a) Forward blocking mode

When anode is positive with respect to cathode, junctions J_1 and J_3 are forward biased. Junction J_2 is reverse biased. A very small leakage current flows from anode to cathode.

b) Forward conduction mode

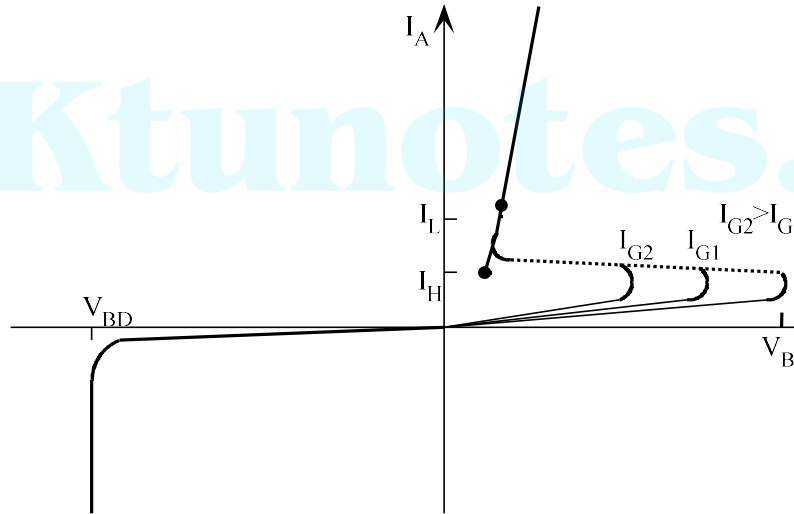
With zero gate current: -

magnitude of gate current. Higher the gate current, lower is the forward breakover voltage.

Once the thyristor is in the on state, the gate cannot be used to turn the device off. The only way to turn off a thyristor is to force the current through the device to be less than the holding current.

Latching current I_L is the minimum anode current required to maintain the on state immediately after it has been turned on and the gate signal has been removed.

Holding current is less than latching current.



c) Reverse blocking mode

When cathode is made positive with respect to anode, the SCR is said to be in reverse blocking mode. In this mode, the junctions J_1 and J_3 are reverse biased, whereas J_2 is forward biased. SCR behaves as if two diodes are connected in series. A small leakage current flows in the SCR. If this reverse current increases, then at a critical breakdown level, called reverse breakdown voltage V_{BD} , avalanche breakdown occurs at J_1 and J_3 and reverse current increases rapidly.

Large current at large voltage V_{BR} leads to increased junction temperature and the device may be damaged.

Static I-V characteristics are shown above.

Note: Generally, the thyristor is designed so that the forward breakover voltage is the same as the reverse breakdown voltage V_{BD} .

For a transistor, $I_C = \alpha I_E + I_{CBO}$

For Q₁, $I_{C1} = \alpha_1 I_A + I_{CBO1}$

For Q₂, $I_{C2} = \alpha_2 I_K + I_{CBO2}$

$$I_A = I_{C1} + I_{B1} = I_{C1} + I_{C2} = \alpha_1 I_A + I_{CBO1} + \alpha_2 I_K + I_{CBO2}$$

But $I_K = I_A + I_G$

$$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)} \dots\dots\dots (1)$$

Current gain α varies with emitter current I_E as shown in figure.

α_1 increases with I_A and α_2 increases with $I_K = I_A + I_G$.

If I_G is increased, I_A increases (equation 1) and $I_K (=I_A + I_G)$ increases. α_1 and α_2 increases; I_A is further increased; there is a regenerative or positive feedback. I_A approaches unity, I_A becomes very large, thyristor is turned ON.

TURN-ON METHODS OF THYRISTOR

1) Forward voltage triggering

When anode to cathode forward voltage is increased with gate circuit open reaches forward breakover voltage (V_{BO}), thyristor starts conducting. The forward current is limited by the load impedance. This method is rarely used as the large V_{BO} may destroy the device. Once the thyristor is conducting, J_2 loses its reverse blocking capability. Hence the forward voltage is reduced below V_{BO} , SCR will continue conduction of the current. Now, the current can be stopped only by reducing the anode current below holding current.

2) Gate triggering

When the thyristor is in forward blocking state and if a positive voltage is applied to the gate-cathode, charges are injected into the inner P layer and the voltage at which conduction occurs is reduced. The forward voltage at which the device switches to ON state depends on the magnitude of gate current. Higher the gate current, lower is the forward breakover voltage.

3) dv/dt triggering

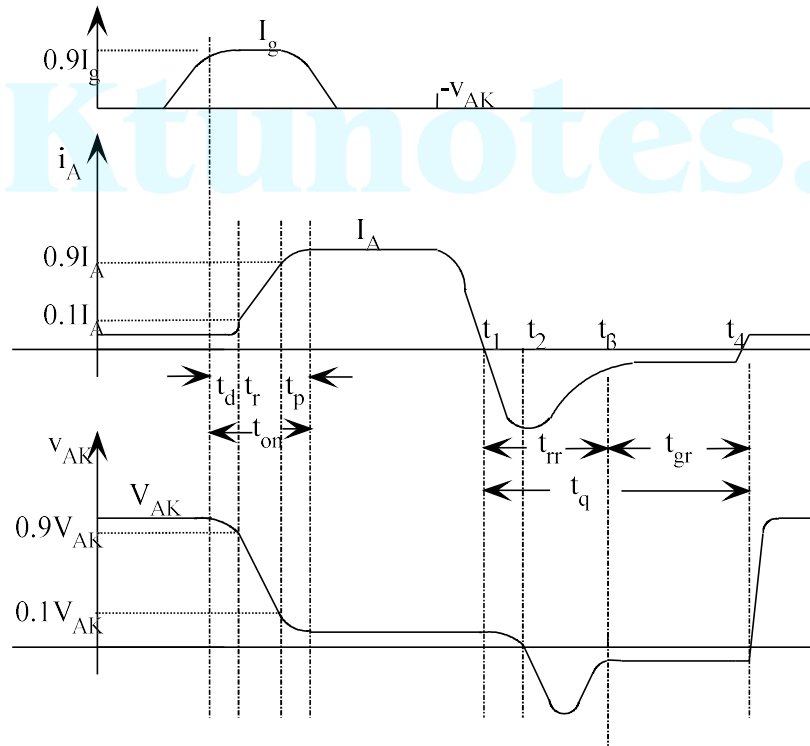
During forward blocking mode, J_1 and J_3 are forward biased and J_2 is reverse biased. Junction J_2 has the characteristics of a capacitor due to charges existing at the junction.

$$i_c = \frac{dQ}{dt} = \frac{d}{dt}(C_j V) = C_j \frac{dV}{dt} + V \frac{dC_j}{dt} \approx C_j \frac{dV}{dt}$$

(rate of change of junction capacitance is negligible as the junction capacitance is very large)

$$i_c = C_j \frac{dV}{dt}$$

time is also defined as the time during which the anode voltage falls from V_A to $0.1V_A$.
 Rise time t_r – Rise time t_r is the time taken by the anode current to rise from $0.1I_A$ to $0.9I_A$, also defined as the time during which the anode voltage falls from $0.9V_A$ to $0.1V_A$.
 Spread time t_p – Spread time is the time taken by the anode current to rise from $0.1I_A$ to $0.9I_A$, also defined as the time during which the anode voltage falls from $0.1V_A$ to $0.9V_A$ (ON state voltage V_{AK} to $2.5V$).



During this time, conduction spreads over the entire cross section of the cathode.

The device switches on very quickly, the turn-on time t_{on} typically being less than $10\mu s$. The width of the gate pulse is in the range 10 to $50\mu s$ and its amplitude in the range 10 to 20V.

TURN OFF CHARACTERISTICS

Once the thyristor is in ON state, the gate has no control on the device. It can be turned off if the anode current becomes zero (natural commutation), or is forced to zero (forced commutation).

At the end of turn-off, a depletion layer develops across junction J_2 and its ability to withstand forward voltage.

di/dt protection

When a thyristor is turned on, conduction starts at a place near the gate. This spreads to the whole area of junction. If the rate of rise of anode current (di/dt) is greater than the spreading velocity of carriers across the cathode junction, local hot spots develop near the gate connection on account of high current density. This causes the junction temperature to rise above the safe limit and as a consequence, SCR may be damaged permanently. To protect against this, an inductance is connected in series with the device (Note that an inductance cannot change instantaneously).

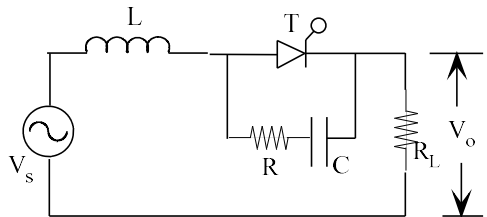
dv/dt protection

During forward blocking state, junction J_2 is reverse biased. Charges across J_2 are C_j . When forward voltage is suddenly applied, a charging current $i_c = C_j \frac{dV}{dt}$ flows.

This current can turn on the SCR even without any gate signal. In order to limit the rate of rise of voltage across the device, a series combination of R and C (RC snubber circuit) is connected in parallel with the thyristor. (Note that voltage across a capacitance cannot change instantaneously).

The snubber circuit is connected to protect the thyristor from unnecessary turn-on. It also reduces transient overvoltage during reverse recovery period and provides a safe operating area in a string of thyristors.

A thyristor with di/dt and dv/dt protection are shown below.



SPECIFICATIONS OF SCR

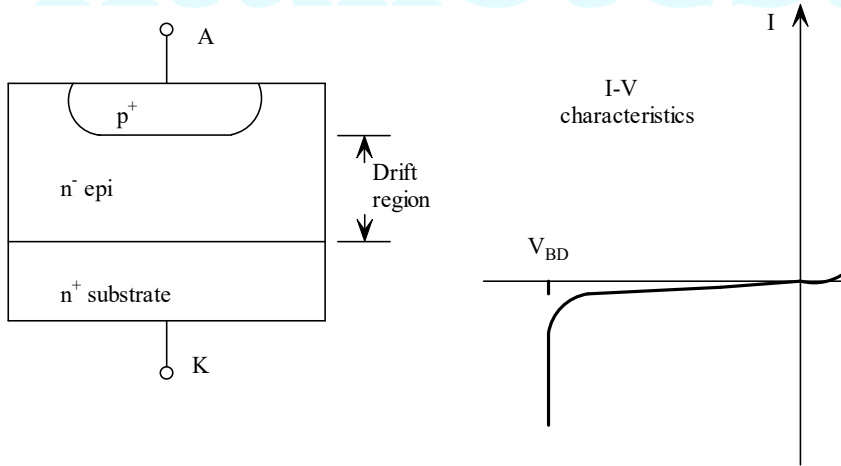
(R-reverse; R-repetitive; M-peak; S-surge; D-forward blocking with gate current; F-forward; T-trigger)

- V_{DWM} – Peak working forward blocking voltage (Maximum forward voltage that the thyristor can withstand during its working)
- V_{DRM} – Peak repetitive forward blocking voltage (peak transient voltage that the thyristor can withstand repeatedly or periodically in its forward blocking mode)

- xi) I_{TSM} – Surge ON-state current (30A for TIC106N)
- xii) I_{GT} – Gate trigger current (200 μ A for TIC106N)
- xiii) V_{GT} – Gate trigger voltage (1V for TIC106N)
- xiv) I_H - Holding current (5mA for TIC106N)
- xv) I^2t rating (choice of fuse or protective equipment for SCR) (4100A²S)
- xvi) di/dt rating (Maximum rate of rise of anode current without any (200A/ μ S)

BASIC STRUCTURE AND PRINCIPLE OF OPERATION POWER DIODE, POWER MOSFET& IGBT

POWER DIODE



n⁻ epitaxial layer (lightly doped) (drift region) is not found in ordinary diode. layer of reverse biased p⁺n⁻ junction.

During reverse bias only a leakage current flows upto V_{BD} where avalanche breakdown occurs; large current flows; large current at large voltage leads to excessive device is destroyed.

Non-punch through diode – width of depletion layer at breakdown is less than depletion layer does not reach n⁺ substrate

Power diode requires finite time to switch from the blocking state to ON state a

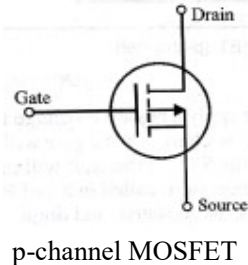
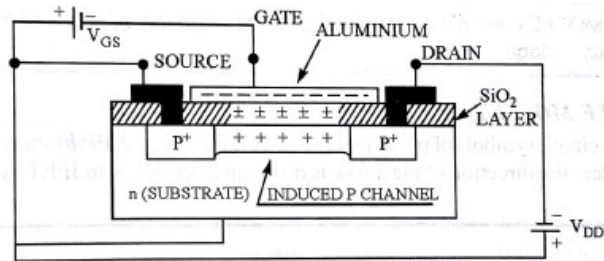
Reverse recovery charge, $q_{rr} = I_{rr} \times \frac{t_{rr}}{2}$

Different types of diodes :

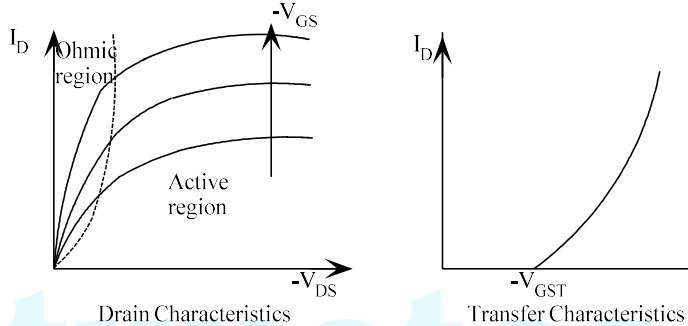
- i) General purpose (rectifier) diodes— have relatively high t_{rr} (25μs) (frequency upto 1kHz); voltage rating upto 6kV
- ii) Fast recovery diodes - low t_{rr} (less than 5μs; used in dc-dc converter frequency switching.
- iii) Schottky diodes – metal to semiconductor junction is used instead of semiconductor junction (aluminium-silicon junction); very low t_{rr} (10ns) used for high frequency applications; available 100V, 400A
- iv) Silicon carbide diodes – SiC is used instead of Si or Ge; negligible t_{rr} behavior; ultra low power loss; very expensive

POWER MOSFET

MOSFET is a 3 terminal device with source S, drain D and gate G as terminals. The current flow from the drain to the source is controlled by the voltage applied between the gate and source. Thus the MOSFET is a voltage controlled device. In the ON-state, the MOSFET allows current to flow in only one direction (from drain to source for n-channel MOSFET and source to drain for p-channel MOSFET). In the OFF state, they can support only unidirectional voltages in both directions like a power transistor.



The structure of a P-channel MOSFET is shown in figure. Two highly doped p+ regions are formed into a lightly doped n-type substrate. One of these p+ regions works as Source and the other as Drain. A thin layer of an insulating material silicon dioxide (SiO₂) is deposited over the substrate. A layer of gate metal, namely Aluminium is deposited over SiO₂ layer, for making connections to the gate, source and drain.



In the drain current V_s gate voltage characteristics (transfer characteristics), when V_{GS} is less than V_{GST} , the drain current is negligible. The drain current starts increasing after a minimum value of V_{GS} . This minimum value of V_{GS} is called the threshold voltage V_{GST} . This V_{GST} is of the order of 1V for power MOSFETs.

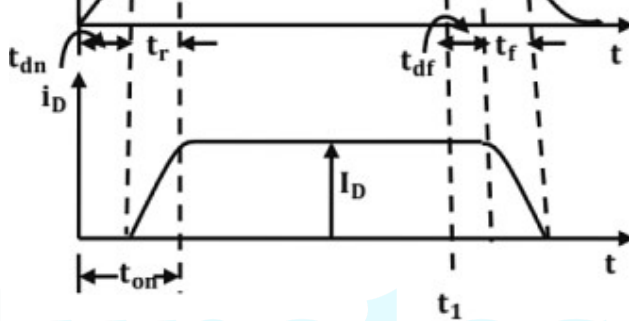
Comparison of BJT and MOSFET

1. BJTs have lower conduction losses in ON state but higher switching losses. MOSFETs have higher conduction losses and lower switching losses. At high frequencies, power MOSFETs are the obvious choice.
2. BJTs are current controlled devices whereas MOSFETs are voltage controlled devices.
3. It is easier to parallel power MOSFETs when compared to BJTs.
4. BJTs suffer from secondary breakdown, but the power MOSFETs do not suffer from secondary breakdown.

SWITCHING CHARACTERISTICS OF POWER MOSFET

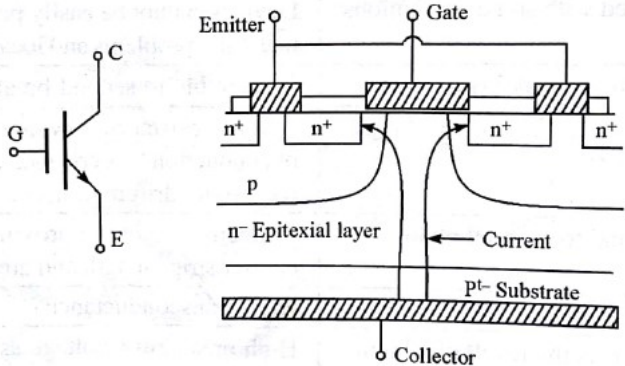
The switching characteristics of a power MOSFET are influenced to a large extent by the input capacitance of the device and the internal impedance of the gate drive circuit. The initial delay t_{dn} during which input capacitance charges to gate threshold voltage is called turn-on delay time.

There is further delay t_r , called rise time, during which gate voltage rises to V_{GS} to drive the MOSFET into on state. During t_r , drain current rises from zero to $I_{D(on)}$. The total turn-on time is $t_{on} = t_d + t_r$. The turn-on time can be reduced by using a high impedance drive source.



INSULATED GATE BIPOLAR TRANSISTOR (IGBT)

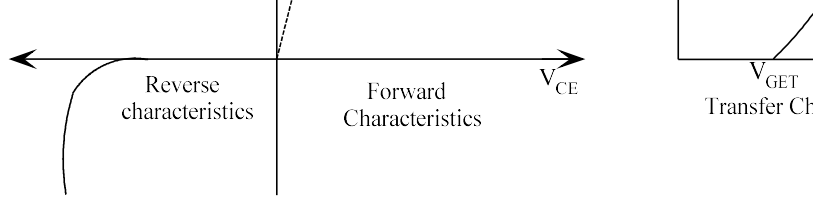
The IGBT is a combination of the advantageous features of MOSFET (high input impedance, voltage control, fast switching) and BJTs (like low ON state loss, high current density, high voltage capability). The IGBT is also a three-terminal device with emitter E, collector C, and gate G as the terminals.



The flow of current from the collector to the emitter is controlled by the voltage applied to the gate-emitter terminals. Therefore, like the MOSFET, the IGBT is also a voltage-controlled device. In the ON-state, the IGBT can allow current in only one direction (from collector to emitter for n-channel IGBT and from emitter to the collector for p-channel IGBT). For the OFF-state, the IGBT can block both forward and reverse bipolar voltages during the OFF state.

The basic structure of an n-channel IGBT is shown in figure. An n-channel IGBT is constructed on a p⁺ substrate. This region constitutes the collector of the IGBT.

When a positive voltage is applied at the collector terminal with respect to emitter and a positive gate-to-emitter voltage, the device can not conduct because the junction between



SWITCHING CHARACTERISTICS OF IGBT

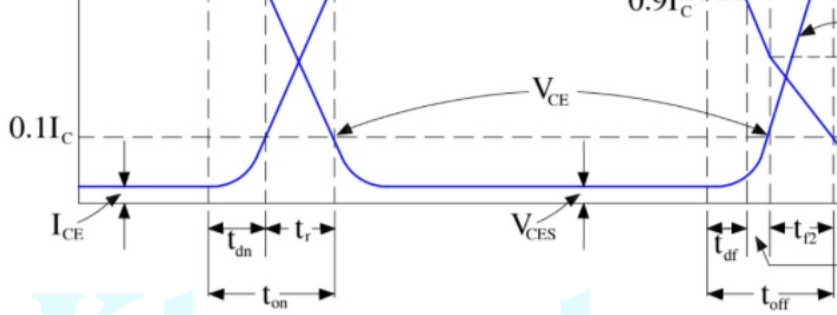
Switching Characteristics of IGBT is basically the graphical representation during its turn-on & turn-off process.

The turn-on time is defined as the time between the instant of forward blocking mode. Turn-on time (t_{on}) is basically composed of two different times: Delay time (t_d) and Rise time (t_r).

$$t_{on} = t_d + t_r.$$

The delay time is defined as the time for the collector-emitter voltage (V_{CE}) to drop to 0.9 V_{CE} . This simply means that, the collector-emitter voltage drops to 90% of its initial value. The collector current rises from initial leakage current to 0.1 I_C (10%). Thus, delay time is defined as the time period during which collector current rises from zero (initial current) to 10% of the final value of collector current I_C .

The rise time t_r is the time during which collector-emitter voltage falls from 90% to 10% of its initial value. This means, during rise time collector-emitter voltage falls to 10% from 90%. The collector current builds up to final value of collector current I_C from 10%. After time t_r , the collector current becomes I_C and the collector-emitter voltage drops to very small value called collector-emitter saturation voltage.



Unlike turn-on time, turn-off time comprises of three intervals: delay time, t_{df} , final fall time, t_{fl} .

Thus, turn-off time is the sum of above three different time intervals i.e. $t_{off} = t_{df} + t_f + t_{fl}$.

The delay time is the time during which gate voltage falls from V_{GE} to threshold voltage. During this time, collector current falls from I_C to $0.9I_C$. After this time, collector-emitter voltage begins to rise.

The first fall time t_{f1} is defined as the time during which collector current falls from I_C to $0.1I_C$. In other words, it is the time during which collector-emitter voltage rises from $0.1V_{CE}$ to V_{CE} .

The final fall time t_{f2} is the time during which collector current falls from $0.1I_C$ to zero. It is the time during which collector-emitter voltage rises from $0.1V_{CE}$ to final value V_{CE} .

Advantages of IGBT are:

Lower gate drive requirements

Low switching losses

Small snubber circuitry requirements

High input impedance

Voltage controlled device

Temperature coefficient of ON state resistance is positive and less than power MOSFET, which results in lower on-state voltage-drop and power loss.

Enhanced conduction due to bipolar nature

Better safe operating area

is a compound of gallium with nitrogen. For this reason, the semiconductor compounds are called “Compound Semiconductors.”

The characterization of a material as being wide bandgap pertains to the electron to jump from the top of the valence band to the bottom of the conduction band of the semiconductor. Materials which require energies typically larger than one eV are referred to as wide bandgap materials. (Si: 1.1 eV, SiC: 3.3 eV, GaN: 3.4 eV)

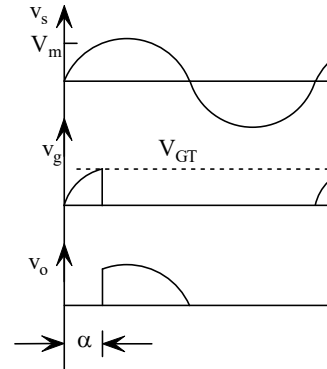
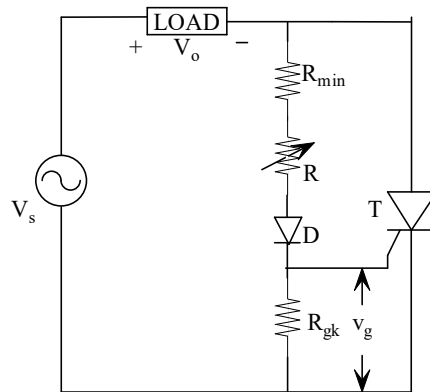
The high critical field of both GaN and SiC compared to Si is a property which allows them to operate at higher voltages and lower leakage currents. Higher electron saturation velocity of GaN and SiC allow for higher frequency of operation. Carrier mobility (specifies the electron speed through the semiconductor material) than Si makes SiC the best device for very high frequencies. SiC has higher thermal conductivity (conducts heat more efficiently) than GaN or Si meaning that SiC devices can theoretically operate at higher power densities than either GaN or Si.

Commercial Si transistors can block voltage upto 400V, GaN transistors upto 600V, SiC transistors upto 1200V.

Compared to silicon, current manufacturing techniques of GaN and SiC are more expensive.

GATE TRIGGERING CIRCUITS

1) RESISTANCE FIRING CIRCUIT



α = firing angle; angle at which the SCR starts conducting

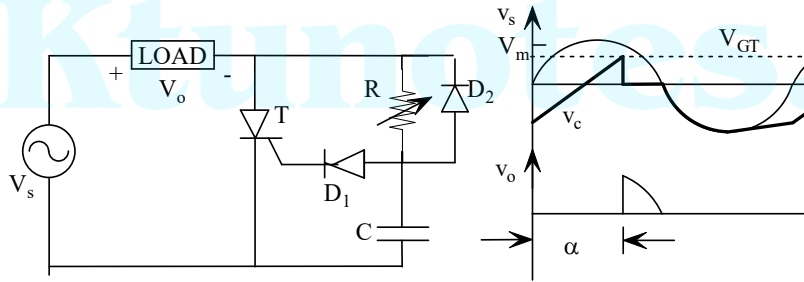
Firing angle can be varied by varying R .

I_{GT} = minimum gate current required to switch the SCR from OFF state to ON state

2) RC FIRING CIRCUIT

Firing angle can be varied from 0 to 180° .

During negative half cycle, capacitor charges to $-V_m$ through diode D_2 with. Capacitor voltage remains constant ($=V_m$) until $v_s=0$. During positive half cycle, capacitor charges through R and when capacitor voltage reaches V_{GT} , the thyristor is triggered.

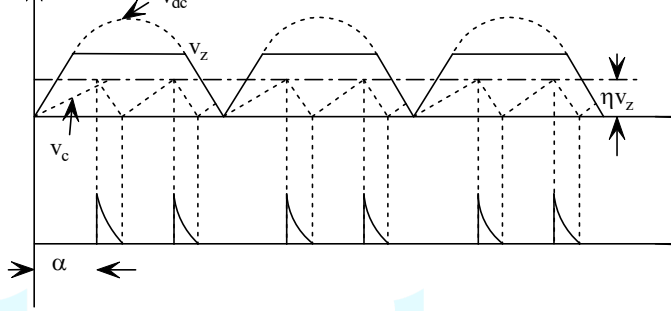


During negative half cycle, diode D_1 prevents the breakdown of the gate to cathode.

Note : For an RC circuit, as R increases, time constant T increases, capacitor charges slowly. Firing angle will be large. As R decreases, T decreases, v_c reaches V_{GT} earlier, firing angle will be less.

3) UJT FIRING CIRCUIT

Diodes D_1 to D_4 are used to rectify ac to dc. Resistor R_1 lowers V_{dc} to a suitable value for the diode and UJT. Zener diode is used to clip the rectified voltage to a voltage V_Z applied to the charging circuit RC. Current i_1 charges capacitor at a rate dV_c/dt . When capacitor voltage v_c reaches the UJT threshold voltage ηV_Z , the E-B₁ junction of the capacitor C discharges through primary of pulse transformer sending a current pulse to the thyristor.



Rate of rise of capacitor voltage can be controlled by varying R . The firing angle can be varied from 0° to about 150° .

COMMUTATION OF SCR

The gate has no control in a SCR after it has been triggered into conduction. To return the SCR to the blocking mode either by reducing the forward current below the holding current or by applying a large reverse voltage across it. This can be achieved by external circuit. This process is called commutation.

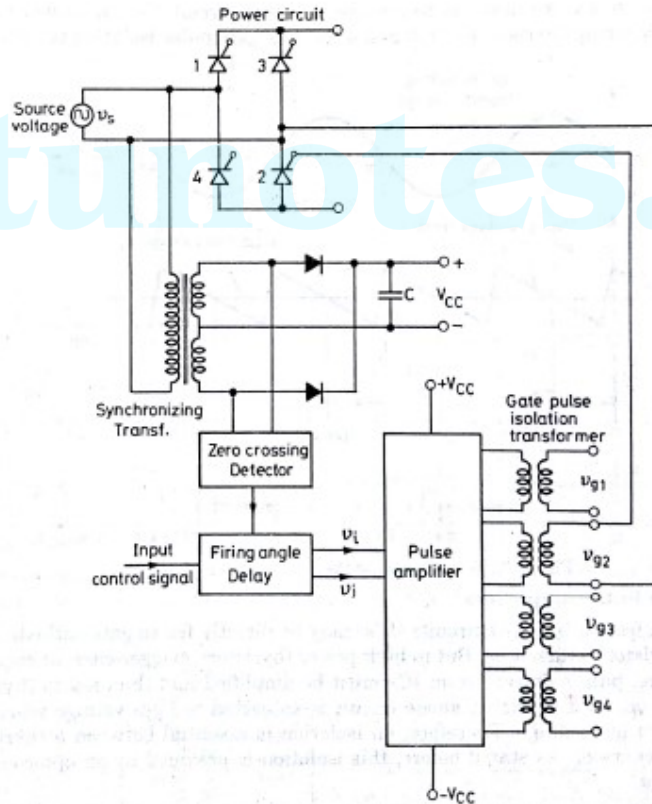
NATURAL OR LINE COMMUTATION

This method is applied for phase-controlled converters, line commutated converters and step down cycloconverters.

The SCR carrying the load current is reverse-biased by the AC source voltage and turns off when anode current falls below the holding current (nearly zero).

In a half-wave controlled rectifier with R -load, SCR is turned ON at firing angle α . From α to π , SCR conducts. At $\omega t = \pi$, current becomes zero and SCR turns off. As the current passes through natural zero, a reverse voltage will simultaneously appear across the device. This process is called as natural commutation since no external circuit is required for this purpose.

- (i) A circuit for the detection of zero crossing of the input voltage
- (ii) Generation of trigger pulses of required waveshape
- (iii) DC power supply for pulse amplifier
- (iv) Gate trigger circuit isolation from the line potential by means of optocouplers



A general block diagram for gate trigger circuit for single-phase controlled rectifier. The gating circuit consists of synchronizing transformer, diode rectifier, zero crossing detector, firing angle delay block, pulse amplifier, gate-pulse isolation transformer and thyristor converter.

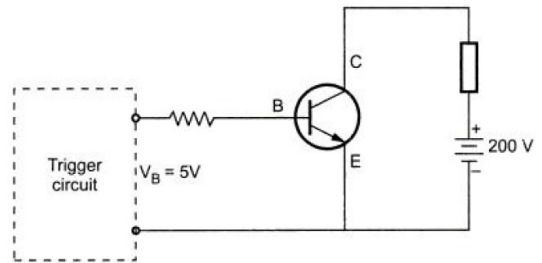
Synchronising transformer steps down the supply voltage suitable for zero crossing detection, delivering dc supply V_{CC} to gate trigger circuit.

with the zero crossing of the ac supply voltage. In the firing angle delay block, ramp voltage is compared with control voltage E_C . When rising ramp voltage E_C , a pulse signal of controlled duration is generated. These signals are indicated by v_i and v_j for thyristors 3 and 4 for the power circuit. If E_C is lowered, firing angle is decreased. If E_C is raised firing angle is increased. This shows that firing delay angle is controlled by the control signal voltage.

The pulse output from the firing delay angle block is fed to a pulse amplifier. The amplified pulses are then used for triggering thyristors 1,2,3 and 4 through gate-pulse isolator.

Necessity of isolation:

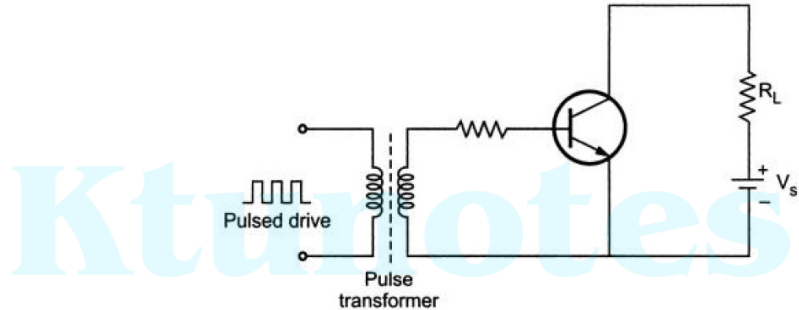
Driver circuits operate at very low power levels. Normally, the signals levels of the driver gate drives are connected to power devices which operate at high power level. This creates a situation.



Observe that the collector of BJT can have voltages of 200V. But base is connected to a trigger circuit that has voltages of 5V. If BJT is damaged and collector-base gets shorted, the trigger circuit will be connected to trigger circuit. This will damage the trigger circuit also. This means the trigger circuit is damaged due to device damage. Therefore, there must be some electrical isolation between the trigger circuit and power circuit. There is one more reason for isolation. Consider that the trigger circuit and the two devices as shown in figure below.

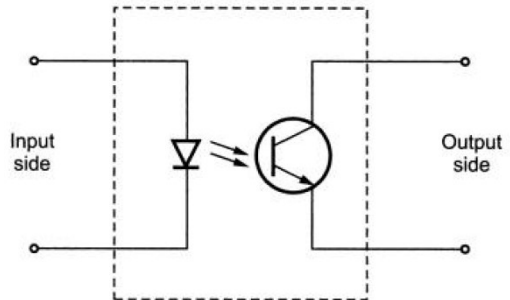
and optocouplers.

Isolation using pulse transformer : Pulse transformer has one primary and one secondary winding. It is normally used for pulsed mode of triggering. Figure shows the pulse transformer.



In the above circuit, the triggering circuit is electrically isolated from BJT. In case of damage to BJT, there will be no effect on triggering circuit. Advantages of pulse transformer does not need external power for its operation ii) it is very simple to use. Disadvantages of pulse transformers are i) pulse transformers saturates at low frequencies hence it can be used at high frequencies ii) due to magnetic coupling, the signal is distorted.

Isolation using optocouplers : Optocoupler consists of a pair of infrared LED and a phototransistor.



When the signal is applied to the infrared LED, it turns on. Its light falls on the phototransistor. Therefore phototransistor also starts conducting. There is no electric connection between the LED and the phototransistor. Figure shows a triggering circuit that uses an optocoupler.

LED turns-on. Its light falls on phototransistor. Hence it turns-on. Therefore, the voltage across the LED goes to zero volts through phototransistor. Due to this, T1 turns-on. Therefore, the voltage across the MOSFET gate becomes zero. Hence MOSFET turns-on. When V_g is zero, the LED turns-off. Hence phototransistor also turns-off. Therefore base drive of T1 goes to VCC and it turns-on. When MOSFET turns-off, MOSFET gate voltage becomes zero. Therefore MOSFET turns-off. Thus the optocoupler works.

Advantages of optoisolators are i) Very good response at low frequencies and ii) cheaper optocoupler devices are available. Disadvantages of optocouplers are i) need for a biasing voltage for its operation and ii) high frequency response is poor.

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